Forced Magnetic Reconnection

1 Preliminary Analysis

Let $\hat{t} = t/(S^{1/3}\,\tau_H)$, where t represents time, S is the Lundquist number, and τ_H the hydromagnetic timescale. Here, S and τ_H are both evaluated at the rational surface. Let $\Psi_b(\hat{t})$ be the driving magnetic flux, and $\Psi_0(\hat{t})$ the resulting reconnected magnetic flux at the rational surface. Let E_{sb} be the (real) dimensionless coupling constant, and $E_{ss} < 0$ the (real) dimensionless tearing stability index.

We can write

$$\bar{\Psi}_b(g) = \int_0^\infty \Psi_b(\hat{t})\,\mathrm{e}^{-g\,\hat{t}}\,d\hat{t}, \tag{1}$$

$$\bar{\Psi}_0(g) = \int_0^\infty \Psi_0(\hat{t})\,\mathrm{e}^{-g\,\hat{t}}\,d\hat{t}. \tag{2}$$

If we assume that

$$\Psi_b(\hat{t}) = \Psi_{b\,0}\left\{\begin{array}{lll} 0 & & \hat{t} < 0 \\ 1 & & \hat{t} \geq 0 \end{array}\right., \tag{3}$$

then

$$\bar{\Psi}_b(g) = \frac{\Psi_{b\,0}}{g}. \tag{4}$$

2 Inner Region

2.1 Layer Equation

The Fourier-Laplace transformed layer equation takes the form

$$\frac{d}{dp}\left(A\,\frac{dY_e}{dp}\right) - \frac{B}{C}\,p^2\,Y_e, \tag{5}$$

where $Y_e(p)$ is the Fourier-Laplace transformed electron stream-function, and

$$A(p) = \frac{p^2}{g + \mathrm{i}\,(Q_E + Q_e) + p^2}, \tag{6}$$

$$
\begin{aligned}
B(p) &= (g + \mathrm{i}\, Q_E)\, [g + \mathrm{i}\, (Q_E + Q_i)] + [g + \mathrm{i}\, (Q_E + Q_i)]\, (P_\varphi + P_\perp)\, p^2 \\
&\quad + P_\varphi\, P_\perp\, p^4, \qquad (7) \\
C(p) &= g + \mathrm{i}\, (Q_E + Q_e) + \{P_\perp + [g + \mathrm{i}\, (Q_E + Q_i)\, D^2]\}\, p^2 \\
&\quad + \iota_e^{-1}\, P_\varphi\, D^2\, p^4. \qquad (8)
\end{aligned}
$$

Here, Q_E, Q_e, and Q_i are the normalized E-cross-B, electron diamagnetic, and ion diamagnetic frequencies, respectively, D is the normalized ion sound radius, and P_φ and $P_\perp$ are the magnetic Prandtl numbers associated with perpendicular momentum and energy transport, respectively. All quantities are evaluated at the rational surface. Moreover, $\iota_e = Q_e/(Q_e - Q_i)$.

The boundary conditions that must be satisfied by a physical solution of Eq. (5) are

$$Y_e(p) = \frac{\hat{\Delta}_s}{\pi\, p} + 1 + \mathcal{O}(p) \qquad (9)$$

as $p \to 0$, and

$$Y_e(p) \to 0 \qquad (10)$$

as $p \to \infty$.

We also need to calculate

$$F_s(g) = -\int_0^\infty A\, \frac{dY_e}{dp}\, dp, \qquad (11)$$

becasue

$$\bar{\Psi}_0(g) = \frac{E_{sb}\, F_s(g)\, \bar{\Psi}_b(g)}{S^{\,1/3}\, \hat{\Delta}_s(g) - E_{ss}} = \frac{\Psi_\infty\, F_s(g)}{g\left[\Sigma\, \hat{\Delta}_s(g) + 1\right]}, \qquad (12)$$

where

$$\Psi_\infty = \frac{E_{sb}\, \Psi_{b\,0}}{(-E_{ss})}, \qquad (13)$$

$$\Sigma = \frac{S^{\,1/3}}{(-E_{ss})}, \qquad (14)$$

and use has been made of Eq. (4).

It is helpful to define

$$\chi(p) = A\, \frac{dY_e}{dp}. \qquad (15)$$

It follows from Eqs. (5) and (11) that

$$A\, \frac{d}{dp}\left(\frac{C}{B\, p^2}\, \frac{d\chi}{dp}\right) - \chi = 0, \qquad (16)$$

and

$$F_s(g) = -\int_0^\infty \chi(p)\, dp. \qquad (17)$$

2.2 Small-p Behavior

Let

$$Y_e(p) = \frac{\hat{\varDelta}_s}{\pi\, p} + 1 + a\,p + b\,p^{\,2} + {\cal O}(p^{\,3}). \tag{18}$$

Substituting into the layer equation, (5), we get

$$\begin{aligned} Y_e(p) &= \frac{\hat{\varDelta}_s}{\pi\, p} + 1 \\ &\quad + \frac{\hat{\varDelta}_s}{\pi}\left\{\frac{1}{2}\,(g+{\rm i}\,Q_E)\,[g+{\rm i}\,(Q_E+Q_i)] - \frac{1}{g+{\rm i}\,(Q_E+Q_e)}\right\} p \\ &\quad + \frac{1}{6}\,(g+{\rm i}\,Q_E)\,[g+{\rm i}\,(Q_E+Q_i)]\,p^{\,2} + {\cal O}(p^{\,3}). \end{aligned} \tag{19}$$

Substituting into Eq. (15), we obtain

$$\begin{aligned} \chi(p) &= -\frac{\hat{\varDelta}_s}{[g+{\rm i}\,(Q_E+Q_e)]\,\pi} + \frac{\varDelta_s}{2\pi}\,\frac{(g+{\rm i}\,Q_E)\,[g+{\rm i}\,(Q_E+Q_i)]}{g+{\rm i}\,(Q_E+Q_e)}\,p^{\,2} \\ &\quad + \frac{1}{3}\,\frac{(g+{\rm i}\,Q_E)\,[g+{\rm i}\,(Q_E+Q_i)]}{g+{\rm i}\,(Q_E+Q_e)}\,p^{\,3} + {\cal O}(p^{\,4}). \end{aligned} \tag{20}$$

2.3 Large-p Behavior

2.3.1 General Case

In the large-p limit, if we write

$$A = 1 + \frac{\alpha}{p^{\,2}}, \tag{21}$$

$$\frac{B}{C} = \beta + \frac{\gamma}{p^{\,2}}, \tag{22}$$

and look for a solution of Eq. (5) of the form

$$Y_e(p) \propto p^{\,x}\,\exp\left(\frac{-\beta^{\,1/2}\,p^{\,2}}{2}\right) \tag{23}$$

then we find that

$$x = \frac{\gamma - \beta^{\,1/2}\,(1-\beta^{\,1/2}\,\alpha)}{2\,\beta^{\,1/2}}. \tag{24}$$

It is easily shown that

$$\alpha = -[g+{\rm i}\,(Q_E+Q_e)], \tag{25}$$

$$\beta = \frac{\iota_e\,P_\perp}{D^{\,2}}, \tag{26}$$

$$\gamma = \frac{\iota_e\,P_\perp}{D^{\,2}}\left(1+[g+{\rm i}\,(Q_E+Q_i)]\,\frac{P_\varphi+P_\perp}{P_\varphi\,P_\perp}-\{P_\perp+[g+{\rm i}\,(Q_E+Q_i)\,D^{\,2}]\}\,\frac{\iota_e}{P_\varphi\,D^{\,2}}\right). \tag{27}$$

Finally, it follows from Eq. (15) that

$$\chi(p) \propto p^{\,x+1}\,\exp\left(\frac{-\beta^{\,1/2}\,p^{\,2}}{2}\right) \tag{28}$$

at large p.

In order to be in the large-p limit, we require

$$p \gg |g+{\rm i}\,(Q_E+Q_e)|^{1/2}, \tag{29}$$

$$p \gg \left|\frac{[g+{\rm i}\,(Q_E+Q_i)]\,(P_\varphi+P_\perp)}{P_\varphi\,P_\perp}\right|^{1/2}, \tag{30}$$

$$p \gg \left|\frac{(g+{\rm i}\,Q_E)\,[g+{\rm i}\,(Q_E+Q_i)]}{P_\varphi\,P_\perp}\right|^{1/4}, \tag{31}$$

$$p \gg \left|\frac{(P_\perp+[g+{\rm i}\,(Q_E+Q_i)\,D^{\,2}])}{\iota_e^{-1}\,P_\varphi\,D^{\,2}}\right|^{1/2}, \tag{32}$$

$$p \gg \left|\frac{g+{\rm i}\,(Q_E+Q_e)}{\iota_e^{-1}\,P_\varphi\,D^{\,2}}\right|^{1/4}, \tag{33}$$

$$p \gg \left(\frac{\iota_e^{-1}\,P_\varphi\,D^{\,2}}{P_\varphi\,P_\perp}\right)^{1/4}. \tag{34}$$

2.3.2 Low-D Limit

If

$$D^{\,2} \ll \left|\frac{P_\perp}{g+{\rm i}\,(Q_E+Q_e)}\right|,\ \frac{\iota_e\,P_\perp}{P_\varphi^{\,2/3}} \tag{35}$$

then the terms in the layer equation involving $D^{\,2}$ are negligible. In this case, in the large-p limit, we can write

$$A = 1+\frac{\tilde{\alpha}}{p^{\,2}}, \tag{36}$$

$$\frac{B}{C} = \tilde{\beta}\,p^{\,2}+\tilde{\gamma}, \tag{37}$$

where

$$\tilde{\alpha} = -[g+{\rm i}\,(Q_E+Q_e)], \tag{38}$$

$$\tilde{\beta} = P_\varphi, \tag{39}$$

$$\tilde{\gamma} = -\mathrm{i}\,(Q_e - Q_i)\,\frac{P_\varphi}{P_\perp} + g + \mathrm{i}\,(Q_E + Q_i). \tag{40}$$

The solution of Eq. (5) becomes

$$Y_e(p) \propto p^{-1}\exp\left(\tilde{x}\,p - \frac{\tilde{\beta}^{\,1/2}\,p^{\,3}}{3}\right) \tag{41}$$

where

$$\tilde{x} = \frac{\tilde{\alpha}\,\tilde{\beta} - \tilde{\gamma}}{2\,\tilde{\beta}^{\,1/2}}. \tag{42}$$

Finally, it is easily seen from Eq. (15) that

$$\chi(p) \propto p\exp\left(\tilde{x}\,p - \frac{\tilde{\beta}^{\,1/2}\,p^{\,3}}{3}\right), \tag{43}$$

at large p.

In order to be in the large-p limit, we require

$$p \gg |g + \mathrm{i}\,(Q_E + Q_e)|^{1/2}, \tag{44}$$

$$p \gg \left|\frac{[g + \mathrm{i}\,(Q_E + Q_i)]\,(P_\varphi + P_\perp)}{P_\varphi\,P_\perp}\right|^{1/2}, \tag{45}$$

$$p \gg \left|\frac{(g + \mathrm{i}\,Q_E)\,[g + \mathrm{i}\,(Q_E + Q_i)]}{P_\varphi\,P_\perp}\right|^{1/4}, \tag{46}$$

$$p \gg \left|\frac{g + \mathrm{i}\,(Q_E + Q_e)}{P_\perp}\right|^{1/2}, \tag{47}$$

$$p \gg P_\varphi^{-1/6}. \tag{48}$$

3 Numerical Solution of Layer Equation

3.1 Ricatti Transformation

Let

$$W(p) = \frac{p}{Y_e}\,\frac{dY_e}{dp}. \tag{49}$$

The layer equation, (5), transforms to give

$$\frac{dW}{dp} = -\frac{A'}{p}\,W - \frac{W^{\,2}}{p} + \frac{B}{A\,C}\,p^{\,3}, \tag{50}$$

where

$$A'(p) = \frac{g + \mathrm{i}\,(Q_E + Q_e) - p^{\,2}}{g + \mathrm{i}\,(Q_E + Q_e) + p^{\,2}}. \tag{51}$$

This equation must be solved subject to the boundary conditions that

$$W(p) = x - \beta^{\,1/2}\,p^{\,2} \tag{52}$$

at large p, and

$$W(p) = -1 + \frac{\pi\,p}{\hat{\Delta}_s} \tag{53}$$

at small p. However, in the low-D limit,

$$W(p) = -1 + \tilde{x}\,p - \tilde{\beta}^{\,1/2}\,p^{\,3} \tag{54}$$

at large p.

Let

$$V(p) = \frac{p}{\chi}\,\frac{d\chi}{dp}. \tag{55}$$

Equation (16) transforms to give

$$\frac{dV}{dp} = 2\,p\,(B' - C')\,V + \frac{3\,V}{p} - \frac{V^2}{p} + \frac{B}{A\,C}\,p^{\,3}, \tag{56}$$

where

$$B'(p) = \frac{[g + {\rm i}\,(Q_E + Q_i)]\,(P_\varphi + P_\perp) + 2\,P_\varphi\,P_\perp\,p^2}{(g + {\rm i}\,Q_E)[g + {\rm i}\,(Q_E + Q_i)] + [g + {\rm i}\,(Q_E + Q_i)]\,(P_\varphi + P_\perp)\,p^{\,2} + P_\varphi\,P_\perp\,p^{\,4}}, \tag{57}$$

$$C'(p) = \frac{\{P_\perp + [g + {\rm i}\,(Q_E + Q_i)\,D^{\,2}\} + 2\,\iota_e^{-1}\,P_\varphi\,D^{\,2}\,p^2}{g + {\rm i}\,(Q_E + Q_e) + \{P_\perp + [g + {\rm i}\,(Q_E + Q_i)\,D^{\,2}\}\,p^{\,2} + \iota_e^{-1}\,P_\varphi\,D^{\,2}\,p^{\,4}}. \tag{58}$$

Equation (56) must be solved subject to the boundary condition that

$$V(p) = 1 + x - \beta^{\,1/2}\,p^{\,2} \tag{59}$$

at large p. However, in the low-D limit,

$$V(p) = 1 + \tilde{x}\,p - \tilde{\beta}^{\,1/2}\,p^{\,3} \tag{60}$$

at large p.

3.2 Method of Solution

3.2.1 Stage 1

First, we launch solutions of Eqs. (50) and (56) from large p, subject to the respective boundary conditions (52) and (59), and integrate them to small p, saving $V(p)$ onto a grid. We then deduce the value of $\hat{\Delta}_s(g)$ from Eq. (53).

3.2.2 Stage 2

Next, we launch solutions of the following system of equations from small p,

$$\frac{dU}{dp} = \frac{V(p)}{p}, \tag{61}$$

$$\frac{dF}{dp} = \exp[U(p)], \tag{62}$$

subject to the boundary conditions

$$U(0) = 0, \tag{63}$$

$$F(0) = 0, \tag{64}$$

and integrate them to large p. Here, $V(p)$ is interpolated from the grid. We then have

$$F_s(g) = \frac{\hat{\Delta}_s}{[g + \mathrm{i}\,(Q_E + Q_e)]\,\pi}\,F(\infty). \tag{65}$$

3.2.3 Stage 3

Finally, we perform the inverse Laplace transform:

$$\hat{\Psi}_0(\hat{t}) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{F_s(\sigma + \mathrm{i}\,\omega)\,\mathrm{e}^{(\sigma+\mathrm{i}\,\omega)\,\hat{t}}\,d\omega}{(\sigma + \mathrm{i}\,\omega)\left[\Sigma\,\hat{\Delta}_s(\sigma + \mathrm{i}\,\omega) + 1\right]}, \tag{66}$$

where $g = \sigma + \mathrm{i}\,\omega$. Here, $\hat{\Psi}_0 = \Psi_0/\Psi_\infty$, ω is real, and σ is real and positive.

4 Linear Response Regimes

4.1 Introduction

The aim of this section is to classify all of the different linear resonant response regimes predicted by Eqs. (5), (9), (10), and (11). In the following, we shall assume that $|Q_E| \sim |Q_e| \sim |Q_i| \sim Q$ and $P_\varphi \sim P_\perp \sim P$.

4.2 Constant-ψ Limit

Suppose that there are two layers in p-space. In the small-p layer, suppose that Eq. (5) reduces to

$$\frac{d}{dp}\left[\frac{p^2}{g + \mathrm{i}\,(Q_E + Q_e) + p^2}\,\frac{dY_e}{dp}\right] \simeq 0 \tag{67}$$

when $p \sim R^{1/2}$, where $R = \sqrt{g^2 + Q^2}$. Integrating directly, making use of Eq. (9), we find that

$$Y_e(p) \simeq \frac{\hat{\Delta}_s}{\pi} \left[\frac{1}{p} - \frac{p}{g + \mathrm{i}\,(Q_E + Q_e)}\right] + 1 + \mathcal{O}(p^2) \tag{68}$$

for $p \lesssim R^{1/2}$.

In the large-p limit, for $|p| \gg \mathcal{O}(R^{1/2})$, Eq. (5) reduces to

$$\frac{d^2 Y_e}{dp^2} - \frac{B}{C}\,p^2\,Y_e \simeq 0, \tag{69}$$

with $Y_e(p)$ bounded as $p \rightarrow \infty$, in accordance with Eq. (10). Asymptotic matching to the small-p layer solution, (68), yields the following small-p boundary condition that the previous equation must satisfy:

$$Y_e(p, g) \simeq 1 - \frac{\hat{\Delta}_s}{\pi}\,\frac{p}{g + \mathrm{i}\,(Q_E + Q_e)} + \mathcal{O}(p^2) \tag{70}$$

as $p \rightarrow 0$.

In the various constant-ψ linear response regimes considered in Sect. 4.3, Eq. (69) reduces to an equation of the form

$$\frac{d^2 Y_e}{dp^2} - G\,p^k\,Y_e \simeq 0, \tag{71}$$

where k is real and non-negative, and $G(g)$ is a complex constant. Let $Y_e(p) = \sqrt{p}\,Z(q)$ and $q = \sqrt{G}\,p^j/j$, where $j = (k+2)/2$. The previous equation transforms into a modified Bessel equation of general order,

$$q^2\,\frac{d^2 Z}{dq^2} + q\,\frac{dZ}{dq} - (q^2 + \nu^2)\,Z = 0, \tag{72}$$

where $\nu = 1/(k+2)$. The solution that is bounded as $q \rightarrow \infty$ has the small-q expansion

$$K_\nu(q) = \frac{1}{\Gamma(1-\nu)}\left(\frac{2}{q}\right)^\nu - \frac{1}{\Gamma(1+\nu)}\left(\frac{q}{2}\right)^\nu + \mathcal{O}\left(q^{\,2-\nu}\right), \tag{73}$$

where $\Gamma(z)$ is a gamma function. A comparison of the previous expression with Eq. (70) reveals that

$$\hat{\Delta}_s(g) = \frac{\nu^{\,2\,\nu-1}\,\pi\,\Gamma(1-\nu)}{\Gamma(\nu)}\,[g + \mathrm{i}\,(Q_E + Q_e)]\,G^\nu. \tag{74}$$

Note, finally, that $p_\ast \sim |G|^{-\nu}$, where $p_\ast$ denotes the width of the large-p layer in p-space. This width must be larger than $R^{1/2}$ (i.e., the width of the small-p layer) in order for the constant-ψ approximation to hold. Finally, it is easily demonstrated from Eqs. (10), (11), (70), and the fact that $p_\ast \gg R^{1/2}$, that

$$F_s(g) \simeq 1 \tag{75}$$

in a constant-ψ layer.

4.3 Constant-ψ Response Regimes

Suppose that $R \gg P\,p^2$ and $D^2\,p^2 \ll 1$. It follows that $k = 2$, $j = 2$, $\nu = 1/4$, and

$$G(g) = \frac{(g + \mathrm{i}\,Q_E)\,[g + \mathrm{i}\,(Q_E + Q_i)]}{g + \mathrm{i}\,(Q_E + Q_e)}. \tag{76}$$

Hence, we deduce that

$$\hat{\Delta}_s(g) = \frac{2\pi\,\Gamma(3/4)}{\Gamma(1/4)}\,[g + \mathrm{i}\,(Q_E + Q_e)]^{3/4}\,(g + \mathrm{i}\,Q_E)^{1/4}\,[g + \mathrm{i}\,(Q_E + Q_i)]^{1/4}. \tag{77}$$

This response regime is known as the *resistive-inertial regime* because the layer response is dominated by plasma resistivity and ion inertia. The characteristic layer width is $p_\ast \sim R^{-1/4}$, which implies that the regime is valid when $P \ll R^{3/2}$, $R \gg D^4$, and $R \ll 1$.

Suppose that $R \ll P\,p^2$ and $D^2\,p^2 \ll 1$. It follows that $k = 4$, $j = 3$, $\nu = 1/6$, and $G(g) = P_\varphi$. Hence, we deduce that

$$\hat{\Delta}_s(g) = \frac{6^{2/3}\,\pi\,\Gamma(5/6)}{\Gamma(1/6)}\,[g + \mathrm{i}\,(Q_E + Q_e)]\,P_\varphi^{1/6}. \tag{78}$$

This response regime is known as the *viscous-resistive regime* because the layer response is dominated by ion perpendicular viscosity and plasma resistivity. The characteristic layer width is $p_\ast \sim P^{-1/6}$, which implies that the regime is valid when $P \gg R^{3/2}$, $P \gg D^6$, and $P \ll R^{-3}$.

Suppose that $R \gg P\,p^2$ and $D^2\,p^2 \gg 1$. It follows that $k = 0$, $j = 1$, $\nu = 1/2$, and

$$G(g) = \frac{g + \mathrm{i}\,Q_E}{D^2}. \tag{79}$$

Hence, we deduce that

$$\hat{\Delta}_s(g) = \frac{\pi\,[g + \mathrm{i}\,(Q_E + Q_e)]\,(g + \mathrm{i}\,Q_E)^{1/2}}{D}. \tag{80}$$

This response regime is known as the *semi-collisional regime*. The characteristic layer width is $p_\ast \sim D\,R^{-1/2}$, which implies that the regime is valid when $R \ll D^4$, $P \ll R^2/D^2$, and $R \ll D$.

Suppose, finally, that $R \ll P\,p^2$ and $D^2\,p^2 \gg 1$. It follows that $k = 2$, $j = 2$, $\nu = 1/4$, and

$$G(g) = \frac{\iota_e\,P_\perp}{D^2}. \tag{81}$$

Hence, we deduce that

$$\hat{\Delta}_s(g) = \frac{2\pi\,\Gamma(3/4)}{\Gamma(1/4)}\,\frac{[g + \mathrm{i}\,(Q_E + Q_e)]\,\iota_e^{1/4}\,P_\perp^{1/4}}{D^{1/2}}. \tag{82}$$

This response regime is known as the *diffusive-resistive regime* because the layer response is dominated by perpendicular energy diffusivity and plasma resistivity. The characteristic layer width is $p_\ast \sim P^{-1/4}\,D^{1/2}$, which implies that the regime is valid when $P \gg R^2/D^2$, $P \ll D^6$, and $P \ll D^2/R^2$.

4.4 Non-Constant-ψ Limit

Suppose that $p \ll R^{1/2}$. In this limit, Eq. (5) reduces to

$$\frac{d}{dp}\left(p^2\,\frac{dY_e}{dp}\right) - [g + \mathrm{i}\,(Q_E + Q_e)]\,\frac{B}{C}\,p^2\,Y_e = 0. \tag{83}$$

In the various non-constant-ψ regimes considered in Sect. 4.5, the previous equation takes the form

$$\frac{d}{dp}\left(p^2\,\frac{dY_e}{dp}\right) - G\,p^{\,k+2}\,Y_e = 0, \tag{84}$$

where k is real and non-negative, and $G(g)$ is a complex constant. Let $V(p) = p\,Y_e(p)$. The previous equation yields

$$\frac{d^2V}{dp^2} - G\,p^k\,V = 0. \tag{85}$$

This equation is identical in form to Eq. (71), which we have already solved. Indeed, the solution that is bounded as $p \to \infty$ has the small-p expansion (73), where $q = \sqrt{G}\,p^j/j$, $j = (k+2)/2$, and $\nu = 1/(k+2)$. Matching to Equation (9) yields

$$\hat{\Delta}_s(g) = -\frac{\pi\,\nu^{\,1-2\,\nu}\,\Gamma(\nu)}{\Gamma(1-\nu)}\,G^{\,-\nu}. \tag{86}$$

The layer width in p-space again scales as $p_\ast \sim |G|^{\,-\nu}$. This width must be less that $R^{1/2}$. Finally, it follows from Eqs. (6) and (11) that

$$F_s(g) = \frac{2}{g + \mathrm{i}\,(Q_E + Q_e)}\int_0^\infty V(p)\,dp \tag{87}$$

in a non-constant-ψ response regime.

4.5 Non-Constant-ψ Linear Resonant Response Regimes

Suppose that $R \gg P\,p^2$ and $D^2\,p^2 \ll 1$. It follows that $k = 0$, $j = 1$, $\nu = 1/2$,

$$G(g) = (g + \mathrm{i}\,Q_E)\,[g + \mathrm{i}\,(Q_E + Q_i)], \tag{88}$$

$$V(p) = -G^{\,-1/2}\,\exp(-G^{\,1/2}\,p). \tag{89}$$

Hence, we deduce that

$$\hat{\Delta}_s(g) = -\frac{\pi}{(g + \mathrm{i}\,Q_E)^{1/2}\,[g + \mathrm{i}\,(Q_E + Q_i)]^{1/2}}, \tag{90}$$

$$F_s(g) = -\frac{2}{[g + \mathrm{i}\,(Q_E + Q_e)]\,(g + \mathrm{i}\,Q_E)\,[g + \mathrm{i}\,(Q_E + Q_i)]}. \tag{91}$$

This response regime is known as the *inertial regime* because the layer response is dominated by ion inertia. The characteristic layer width is $p_\ast \sim R^{-1}$, which implies that the regime is valid when $P \ll R^3$, $R \gg D$, and $R \gg 1$.

Suppose that $R \ll P\,p^2$ and $D^2\,p^2 \ll 1$. It follows that $k=2$, $j=2$, $\nu = 1/4$, and

$$G(g) = [g + {\rm i}\,(Q_E+Q_e)]\,P_\varphi, \tag{92}$$

$$V(p) = -\frac{\Gamma(1/4)}{\sqrt{\pi}\,2^{3/4}\,G^{1/4}}\,U(0, 2^{1/2}\,G^{1/4}\,p), \tag{93}$$

where $U(a,x) = D_{-a-1/2}(x)$ is a parabolic cylinder function. Hence, we deduce that

$$\hat{\Delta}_s(g) = -\frac{\pi}{2}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}\,\frac{1}{[g-{\rm i}\,(Q_E+Q_e)]^{1/4}\,P_\varphi^{1/4}}, \tag{94}$$

$$F_s(g) = -\frac{\pi^{1/2}}{2}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}\,\frac{1}{[g-{\rm i}\,(Q_E+Q_e)]^{3/2}\,P_\varphi^{1/2}}, \tag{95}$$

where use has been made of the result

$$\int_0^\infty U(0,x)\,dx = \frac{\pi}{2^{3/4}\,\Gamma(3/4)}. \tag{96}$$

This response regime is known as the *viscous-inertial regime* because the layer response is dominated by ion perpendicular viscosity and ion inertia. The characteristic layer width is $p_\ast \sim R^{-1/4}\,P^{-1/4}$, which implies that the regime is valid when $P \gg R^3$, $P \gg D^4/R$, and $P \gg R^{-3}$.

Suppose, finally, that $R \ll P\,p^2$ and $D^2\,p^2 \gg 1$. It follows that $k=0$, $j=1$, $\nu = 1/2$, and

$$G(g) = \frac{[g+{\rm i}\,(Q_E+Q_e)]\,\iota_e\,P_\perp}{D^2}, \tag{97}$$

$$V(p) = -G^{-1/2}\,\exp(-G^{1/2}\,p). \tag{98}$$

Hence, we deduce that

$$\hat{\Delta}_s = -\frac{\pi\,D}{[g+{\rm i}\,(Q_E+Q_e)]^{1/2}\,\iota_e^{1/2}\,P_\perp^{1/2}}, \tag{99}$$

$$F_s(g) = -\frac{2\,D^2}{[g+{\rm i}\,(Q_E+Q_e)]^2\,\iota_e\,P_\perp}. \tag{100}$$

This response regime is known as the *diffusive-inertial regime* because the layer response is dominated by perpendicular energy diffusivity and ion inertia. The characteristic layer width is $p_\ast \sim R^{-1/2}\,P^{-1/2}\,D$, which implies that the regime is valid when $R \ll D$, $P \ll D^4/Q$, and $P \gg D^2/R^2$.

4.6 Linear Resonant Response Regimes

Figures 1 and 2 show the extents of the various different linear resonant response regimes in R-P space for the cases $D < 1$ and $D > 1$, respectively.

5 Inverse Laplace Transforms

5.1 Introduction

We can write

$$\hat{\Psi}_0(\hat{t}) = \frac{1}{2\pi\,\mathrm{i}} \int_C \tilde{\Psi}_0(g)\,\mathrm{e}^{\,g\,\hat{t}}\,dg, \tag{101}$$

where C is the Bromwich contour, and

$$\tilde{\Psi}_0(g) = \frac{F_s(g)}{g\left[\Sigma\,\hat{\Delta}_s(g) + 1\right]}. \tag{102}$$

In the various non-constant-ψ regimes, we shall assume that $\Sigma\,|\Delta_s(g)| \gg 1$, so

$$\tilde{\Psi}_0(g) \simeq \frac{F_s(g)}{\Sigma\,g\,\hat{\Delta}_s(g)}. \tag{103}$$

In the various constant-ψ regimes, $F_s \simeq 1$, so

$$\tilde{\Psi}_0(g) = \frac{1}{g\left[\Sigma\,\hat{\Delta}_s(g) + 1\right]}. \tag{104}$$

5.2 Inertial Regime

In the inertial regime,

$$\hat{\Delta}_s(g) = -\frac{\pi}{(g + \mathrm{i}\,Q_E)^{1/2}\,[g + \mathrm{i}\,(Q_E + Q_i)]^{1/2}}, \tag{105}$$

$$F_s(g) = -\frac{2}{[g + \mathrm{i}\,(Q_E + Q_e)]\,(g + \mathrm{i}\,Q_E)\,[g + \mathrm{i}\,(Q_E + Q_i)]}, \tag{106}$$

$$\frac{F_s(g)}{\hat{\Delta}_s(g)} = \frac{2}{\pi}\,\frac{1}{[g + \mathrm{i}\,(Q_E + Q_e)]\,(g + \mathrm{i}\,Q_E)^{1/2}\,[g + \mathrm{i}\,(Q_E + Q_i)]^{1/2}} \tag{107}$$

Thus,

$$\tilde{\Psi}_{0\,I}(g) = \frac{\epsilon_I}{\mathrm{i}\,(Q_E + Q_e)}\left[\frac{1}{g} - \frac{1}{g + \mathrm{i}\,(Q_E + Q_e)}\right]\frac{1}{(g + \mathrm{i}\,Q_E)^{1/2}\,[g + \mathrm{i}\,(Q_E + Q_i)]^{1/2}}, \tag{108}$$

where

$$\epsilon_I = \frac{2}{\pi\,\Sigma}. \tag{109}$$

Now, if

$$\tilde{\Psi}_0(g) = \frac{\bar{H}(g)}{g+a} \tag{110}$$

then (ERD:[1] 4.1.20)

$$\hat{\Psi}_0(\hat{t}) = \int_0^{\hat{t}} {\rm e}^{\,a\,(\hat{t}'-\hat{t})}\,H(\hat{t}')\,d\hat{t}', \tag{111}$$

where $\bar{H}(g)$ is the Laplace transform of $H(\hat{t})$. In this case,

$$\bar{H}(g) = \frac{1}{(g+{\rm i}\,Q_E)^{1/2}\,[g+{\rm i}\,(Q_E+Q_i)]^{1/2}} = \frac{1}{(g^{\,2}+\alpha\,g+\beta)^{1/2}}, \tag{112}$$

where

$$\alpha = {\rm i}\,(2\,Q_E+Q_i), \tag{113}$$

$$\beta = -Q_E\,(Q_E+Q_i). \tag{114}$$

Note that

$$\beta - \frac{1}{4}\,\alpha^2 = -Q_E\,(Q_E+Q_i) + \frac{1}{4}\,(2\,Q_E+Q_i)^2 = \frac{Q_i^2}{4}. \tag{115}$$

However (ERD: 5.3.34),

$$H(\hat{t}) = {\rm e}^{-\alpha\,\hat{t}/2}\,J_0[(\beta-\alpha^2/4)^{1/2}\,\hat{t}] = {\rm e}^{-{\rm i}\,(Q_E+Q_i/2)\,\hat{t}}\,J_0\left(\frac{Q_i}{2}\,\hat{t}\right). \tag{116}$$

Thus,

$$\hat{\Psi}_{0\,I}(\hat{t}) = \frac{\epsilon_I}{{\rm i}\,(Q_E+Q_e)}\int_0^{\hat{t}}\left[1-{\rm e}^{\,{\rm i}\,(Q_E+Q_e)\,(\hat{t}'-\hat{t})}\right]{\rm e}^{-{\rm i}\,(Q_E+Q_i/2)\,\hat{t}'}\,J_0\left(\frac{Q_i}{2}\,\hat{t}'\right)d\hat{t}'. \tag{117}$$

At small $\hat{t}$,

$$\hat{\Psi}_{0\,I}(\hat{t}) \simeq \frac{\epsilon_I\,\hat{t}^{\,2}}{\Gamma(3)}. \tag{118}$$

At large $\hat{t}$,

$$\hat{\Psi}_{0\,I}(\hat{t}) \simeq \frac{\epsilon_I}{{\rm i}\,(Q_E+Q_e)}\int_0^{\infty}\left[1-{\rm e}^{\,{\rm i}\,(Q_E+Q_e)\,(\hat{t}'-\hat{t})}\right]{\rm e}^{-{\rm i}\,(Q_E+Q_i/2)\,\hat{t}'}\,J_0\left(\frac{Q_i}{2}\,\hat{t}'\right)d\hat{t}'. \tag{119}$$

[1] Erdélyi, et al. *Tables of Integral Transforms*, vol. I.

But (GR:[2] 6.621.4)

$$\int_0^\infty \mathrm{e}^{-\alpha\,x}\,J_0(\beta\,x)\,dx = \frac{1}{\sqrt{\alpha^2+\beta^2}}, \tag{120}$$

so

$$\hat{\Psi}_{0\,I}(\hat{t}) \simeq \frac{\epsilon_I}{\mathrm{i}\,(Q_E+Q_i)}\left\{\frac{1}{(\mathrm{i}\,Q_E)^{1/2}\,[\mathrm{i}\,(Q_E+Q_i)]^{1/2}} - \frac{\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}}}{(\mathrm{i}\,Q_e)^{1/2}\,[\mathrm{i}\,(Q_e-Q_i)]^{1/2}}\right\}. \tag{121}$$

5.3 Viscous-Inertial Regime

In the viscous-inertial regime,

$$\hat{\Delta}_s(g) = -\frac{\pi}{2}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}\,\frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{1/4}\,P_\varphi^{1/4}}, \tag{122}$$

$$F_s(g) = -\frac{\pi^{1/2}}{2}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}\,\frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{3/2}\,P_\varphi^{1/2}}, \tag{123}$$

$$\frac{F_s(g)}{\hat{\Delta}_s(g)} = \frac{1}{\pi^{1/2}}\,\frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{5/4}\,P_\varphi^{1/4}}. \tag{124}$$

Thus,

$$\tilde{\Psi}_{0\,VI}(g) = \frac{\epsilon_{VI}}{g}\,\frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{5/4}}, \tag{125}$$

where

$$\epsilon_{VI} = \frac{1}{\pi^{\,1/2}\,\Sigma\,P_\varphi^{1/4}}. \tag{126}$$

In this case,

$$\bar{H}(g) = \frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{5/4}}, \tag{127}$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(5/4)}\,\hat{t}^{\,1/4}\,\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}}. \tag{128}$$

Hence,

$$\hat{\Psi}_{0\,VI}(\hat{t}) = \frac{\epsilon_{VI}}{\Gamma(5/4)}\int_0^{\hat{t}} \hat{t}'^{\,1/4}\,\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}' \tag{129}$$

At small $\hat{t}$,

$$\hat{\Psi}_{0\,VI}(\hat{t}) \simeq \frac{\epsilon_{VI}\,\hat{t}^{\,5/4}}{\Gamma(9/4)}. \tag{130}$$

[2] I.S. Gradshteyn and I.M. Ryzhik, *Table of Integrals, Series, and Products*, Corrected and Enlarged Edition

At large $\hat{t}$,

$$\hat{\Psi}_{0\,VI}(\hat{t}) \simeq \frac{\epsilon_{VI}}{\Gamma(5/4)} \int_0^\infty \hat{t}'^{\,1/4}\, \mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}'}\, d\hat{t}' \tag{131}$$

But (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1}\, \mathrm{e}^{-\mathrm{i}\,q\,x}\, dx = \frac{\Gamma(\nu)}{(\mathrm{i}\,q)^\nu}, \tag{132}$$

so

$$\hat{\Psi}_{0\,VI}(\hat{t}) \simeq \frac{\epsilon_{VI}}{[\mathrm{i}\,(Q_E+Q_e)]^{5/4}}. \tag{133}$$

5.4 Diffusive-lnertial Regime

In the diffusive-inertial regime,

$$\hat{\Delta}_s(g) = -\frac{\pi\,D}{[g+\mathrm{i}\,(Q_E+Q_e)]^{1/2}\,\iota_e^{1/2}\,P_\perp^{1/2}}, \tag{134}$$

$$F_s(g) = -\frac{2\,D^2}{[g+\mathrm{i}\,(Q_E+Q_e)]^2\,\iota_e\,P_\perp}, \tag{135}$$

$$\frac{F_s(g)}{\hat{\Delta}_s(g)} = \frac{2}{\pi}\,\frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{3/2}\,\iota_e^{1/2}\,P_\perp^{1/2}\,D^{-1}}. \tag{136}$$

Thus,

$$\tilde{\Psi}_{0\,DI}(g) = \frac{\epsilon_{DI}}{g}\,\frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{3/2}}, \tag{137}$$

where

$$\epsilon_{DI} = \frac{2}{\pi}\,\frac{1}{\Sigma\,\iota_e^{1/2}\,P_\perp^{1/2}\,D^{-1}}. \tag{138}$$

In this case,

$$\bar{H}(g) = \frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{3/2}}, \tag{139}$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(3/2)}\,\hat{t}^{\,1/2}\,\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}}. \tag{140}$$

Hence,

$$\hat{\Psi}_{0\,DI}(\hat{t}) = \frac{\epsilon_{DI}}{\Gamma(3/2)} \int_0^{\hat{t}} \hat{t}'^{\,1/2}\, \mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}'}\, d\hat{t}' \tag{141}$$

At small $\hat{t}$,

$$\hat{\Psi}_{0\,DI}(\hat{t}) \simeq \frac{\epsilon_{DI}\,\hat{t}^{\,3/2}}{\Gamma(5/2)}. \tag{142}$$

At large $\hat{t}$,

$$\hat{\Psi}_{0\,DI}(\hat{t}) = \frac{\epsilon_{DI}}{\Gamma(3/2)}\int_0^\infty \hat{t}'^{\,1/2}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}' \tag{143}$$

But (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1}\,{\rm e}^{-{\rm i}\,q\,x}\,dx = \frac{\Gamma(\nu)}{({\rm i}\,q)^\nu}, \tag{144}$$

so

$$\hat{\Psi}_{0\,DI}(\hat{t}) \simeq \frac{\epsilon_{DI}}{[{\rm i}\,(Q_E+Q_e)]^{3/2}}. \tag{145}$$

5.5 Viscous-Resistive Regime

In the viscous-resistive regime,

$$\hat{\Delta}_s(g) = \frac{6^{2/3}\,\pi\,\Gamma(5/6)}{\Gamma(1/6)}\,[g+{\rm i}\,(Q_E+Q_e)]\,P_\varphi^{1/6}, \tag{146}$$

so

$$\tilde{\Psi}_{0\,VR}(g) = \frac{\epsilon_{VR}}{g}\,\frac{1}{[g+{\rm i}\,(Q_E+Q_e)]+\epsilon_{VR}}, \tag{147}$$

where

$$\epsilon_{VR} = \frac{\Gamma(1/6)}{6^{2/3}\,\pi\,\Gamma(5/6)}\,\frac{1}{\Sigma\,P_\varphi^{1/6}}. \tag{148}$$

Thus,

$$\tilde{\Psi}_{0\,VR}(g) = \frac{\epsilon_{VR}}{{\rm i}\,(Q_E+Q_e)+\epsilon_{VR}}\left(\frac{1}{g}-\frac{1}{g+[{\rm i}\,(Q_E+Q_e)+\epsilon_{VR}]}\right), \tag{149}$$

giving

$$\hat{\Psi}_{0\,VR}(\hat{t}) = \frac{\epsilon_{VR}}{{\rm i}\,(Q_E+Q_e)+\epsilon_{VR}}\left(1-{\rm e}^{-[{\rm i}\,(Q_E+Q_e)+\epsilon_{VR}]\,\hat{t}}\right). \tag{150}$$

At small $\hat{t}$,

$$\hat{\Psi}_{0\,VR}(\hat{t}) \simeq \frac{\epsilon_{VR}\,\hat{t}}{\Gamma(2)}. \tag{151}$$

At large $\hat{t}$,

$$\hat{\Psi}_{0\,VR}(\hat{t}) \simeq \frac{\epsilon_{VR}}{{\rm i}\,(Q_E+Q_e)+\epsilon_{VR}}. \tag{152}$$

5.6 Diffusive-Resistive Regime

In the diffusive-resistive regime,

$$\hat{\Delta}_s(g) = \frac{2\pi\,\Gamma(3/4)}{\Gamma(1/4)}\,[g + \mathrm{i}\,(Q_E+Q_e)]\,\iota_e^{1/4}\,P_\perp^{1/4}\,D^{-1/2}, \tag{153}$$

so

$$\tilde{\Psi}_{0\,DR}(g) = \frac{\epsilon_{DR}}{g}\,\frac{1}{[g + \mathrm{i}\,(Q_E+Q_e)] + \epsilon_{DR}}, \tag{154}$$

where

$$\epsilon_{DR} = \frac{\Gamma(1/4)}{2\pi\,\Gamma(3/4)}\,\frac{1}{\Sigma\,\iota_e^{1/4}\,P_\perp^{1/4}\,D^{-1/2}}. \tag{155}$$

Thus,

$$\tilde{\Psi}_{0\,DR}(g) = \frac{\epsilon_{DR}}{\mathrm{i}\,(Q_E+Q_e)+\epsilon_{DR}}\left(\frac{1}{g} - \frac{1}{g + [\mathrm{i}\,(Q_E+Q_e)+\epsilon_{DR}]}\right), \tag{156}$$

giving

$$\hat{\Psi}_{0\,DR}(\hat{t}) = \frac{\epsilon_{DR}}{\mathrm{i}\,(Q_E+Q_e)+\epsilon_{DR}}\left(1 - \mathrm{e}^{-[\mathrm{i}\,(Q_E+Q_e)+\epsilon_{DR}]\,\hat{t}}\right). \tag{157}$$

At small $\hat{t}$,

$$\Psi_{0\,DR}(\hat{t}) \simeq \frac{\epsilon_{DR}\,\hat{t}}{\Gamma(2)}. \tag{158}$$

At large $\hat{t}$,

$$\hat{\Psi}_{0\,DR}(\hat{t}) \simeq \frac{\epsilon_{DR}}{\mathrm{i}\,(Q_E+Q_e)+\epsilon_{DR}}. \tag{159}$$

5.7 Semi-Collisional Regime

In the semi-collisional regime,

$$\hat{\Delta}_s(g) = \pi\,[g + \mathrm{i}\,(Q_E+Q_e)]\,(g+\mathrm{i}\,Q_E)^{1/2}\,D^{-1}, \tag{160}$$

so

$$\tilde{\Psi}_{0\,SC}(g) = \frac{\epsilon_{SC}}{g}\,\frac{1}{[g + \mathrm{i}\,(Q_E+Q_e)]\,(g+\mathrm{i}\,Q_E)^{1/2} + \epsilon_{SC}}, \tag{161}$$

where

$$\epsilon_{SC} = \frac{1}{\pi\,\Sigma\,D^{-1}}. \tag{162}$$

If we neglect ϵ_{SC} in the denominator then

$$\tilde{\Psi}_{0\,SC}(g) \simeq \frac{\epsilon_{SC}}{g}\,\frac{1}{[g + \mathrm{i}\,(Q_E+Q_e)]\,(g+\mathrm{i}\,Q_E)^{1/2}}, \tag{163}$$

giving

$$\tilde{\Psi}_{0\,SC}(g) \simeq \frac{\epsilon_{SC}}{\mathrm{i}\,(Q_E+Q_e)}\left[\frac{1}{g}-\frac{1}{g+\mathrm{i}\,(Q_E+Q_e)}\right]\frac{1}{(g+\mathrm{i}\,Q_E)^{1/2}}. \tag{164}$$

In this case,

$$\bar{H}(g) = \frac{1}{(g+\mathrm{i}\,Q_E)^{1/2}}, \tag{165}$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(1/2)}\,\hat{t}^{-1/2}\,\mathrm{e}^{-\mathrm{i}\,Q_E\,\hat{t}}. \tag{166}$$

Hence,

$$\hat{\Psi}_{0\,SC}(\hat{t}) = \frac{\epsilon_{SC}}{\Gamma(1/2)\,\mathrm{i}\,(Q_E+Q_e)}\int_0^{\hat{t}}\left[1-\mathrm{e}^{\mathrm{i}\,(Q_E+Q_e)\,(\hat{t}'-\hat{t})}\right]\hat{t}'^{-1/2}\,\mathrm{e}^{-\mathrm{i}\,Q_E\,\hat{t}'}\,d\hat{t}'. \tag{167}$$

At small $\hat{t}$,

$$\hat{\Psi}_{0\,SC}(\hat{t}) = \frac{\epsilon_{SC}\,\hat{t}^{\,3/2}}{\Gamma(5/2)}. \tag{168}$$

At large $\hat{t}$,

$$\hat{\Psi}_{0\,SC}(\hat{t}) = \frac{\epsilon_{SC}}{\Gamma(1/2)\,\mathrm{i}\,(Q_E+Q_e)}\int_0^{\infty}\left[1-\mathrm{e}^{\mathrm{i}\,(Q_E+Q_e)\,(\hat{t}'-\hat{t})}\right]\hat{t}'^{-1/2}\,\mathrm{e}^{-\mathrm{i}\,Q_E\,\hat{t}'}\,d\hat{t}'. \tag{169}$$

But (GR: 3.381.5),

$$\int_0^{\infty} x^{\nu-1}\,\mathrm{e}^{-\mathrm{i}\,q\,x}\,dx = \frac{\Gamma(\nu)}{(\mathrm{i}\,q)^{\nu}}, \tag{170}$$

It follows that

$$\hat{\Psi}_{0\,SC}(\hat{t}) = \frac{\epsilon_{SC}}{\mathrm{i}\,(Q_E+Q_e)}\left[\frac{1}{(\mathrm{i}\,Q_E)^{1/2}}-\frac{\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}}}{(-\mathrm{i}\,Q_e)^{1/2}}\right]. \tag{171}$$

5.8 Resistive-Inertial Regime

In the resistive-inertial regime,

$$\hat{\Delta}_s(g) = \frac{2\pi\,\Gamma(3/4)}{\Gamma(1/4)}\,[g+\mathrm{i}\,(Q_E+Q_e)]^{3/4}\,(g+\mathrm{i}\,Q_E)^{1/4}\,[g+\mathrm{i}\,(Q_E+Q_i)]^{1/4}, \tag{172}$$

so

$$\tilde{\Psi}_{0\,RI}(g) = \frac{\epsilon_{RI}}{g}\,\frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{3/4}\,(g+\mathrm{i}\,Q_E)^{1/4}\,[g+\mathrm{i}\,(Q_E+Q_i)]^{1/4}+\epsilon_{RI}}, \tag{173}$$

where

$$\epsilon_{RI} = \frac{\Gamma(1/4)}{2\pi\,\Gamma(3/4)\,\Sigma}. \tag{174}$$

If we neglect ϵ_{RI} in the denominator then

$$\tilde{\Psi}_{0\,RI}(g) \simeq \frac{\epsilon_{RI}}{g}\,\frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{3/4}\,(g+\mathrm{i}\,Q_E)^{1/4}\,[g+\mathrm{i}\,(Q_E+Q_i)]^{1/4}}. \tag{175}$$

Now, if

$$\tilde{\Psi}_{0\,RI}(g) \simeq \epsilon_{RI}\,\frac{H(g)}{g} \tag{176}$$

then

$$\hat{\Psi}_{0\,RI}(\hat{t}) = \epsilon_{RI}\int_0^{\hat{t}} H(\hat{t}')\,d\hat{t}'. \tag{177}$$

Furthermore, if

$$\bar{H}(g) = \bar{H}_1(g)\,\bar{H}_2(g) \tag{178}$$

then

$$H(\hat{t}) = \int_0^{\hat{t}} H_1(\hat{t}')\,H_2(\hat{t}-\hat{t}')\,d\hat{t}'. \tag{179}$$

It follows that

$$\hat{\Psi}_{0\,RI}(\hat{t}) = \epsilon_{RI}\int_0^{\hat{t}}\left[\int_0^{\hat{t}'} H_1(\hat{t}'')\,H_2(\hat{t}'-\hat{t}'')\,d\hat{t}''\right]d\hat{t}'. \tag{180}$$

In the present case

$$\bar{H}_1(g) = \frac{1}{(g+\mathrm{i}\,Q_E)^{1/4}\,[g+\mathrm{i}\,(Q_E+Q_i)]^{1/4}}, \tag{181}$$

$$\bar{H}_2(g) = \frac{1}{[g+\mathrm{i}\,(Q_E+Q_e)]^{3/4}}, \tag{182}$$

so (ERD: 5.4.5, AS:[3] 9.6.3, ERD: 5.4.1)

$$H_1(\hat{t}) = \frac{\pi^{1/2}}{\Gamma(1/4)}\,Q_i^{1/4}\,\hat{t}^{-1/4}\,\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_i/2)\,\hat{t}}\,J_{-1/4}\left(\frac{Q_i}{2}\,\hat{t}\right), \tag{183}$$

$$H_2(\hat{t}) = \frac{1}{\Gamma(3/4)}\,\hat{t}^{-1/4}\,\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}}. \tag{184}$$

Note that

$$\Gamma(1/4)\,\Gamma(3/4) = \sqrt{2}\,\pi. \tag{185}$$

Thus,

$$\hat{\Psi}_{0\,RI}(\hat{t}) = \frac{\epsilon_{RI}\,Q_i^{1/4}}{\sqrt{2\pi}}\int_0^{\hat{t}}\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}'}\left[\int_0^{\hat{t}'}(\hat{t}'-\hat{t}'')^{-1/4}\,\hat{t}''^{-1/4}\mathrm{e}^{\,\mathrm{i}\,(Q_e-Q_i/2)\,\hat{t}''}\,J_{-1/4}\left(\frac{Q_i}{2}\,\hat{t}''\right)d\hat{t}''\right]d\hat{t}'. \tag{186}$$

[3]M. Abramowitz and I.A. Stegun, *Handbook of Mathematical Functions* (Dover, New York NY, 1964).

At small $\hat{t}$ (AS: 9.1.10),

$$J_{-1/4}\left(\frac{Q_i}{2}\,\hat{t}\right)\simeq\frac{\sqrt{2}\,Q_i^{-1/4}\,\hat{t}^{-1/4}}{\Gamma(3/4)},\tag{187}$$

so

$$\hat{\Psi}_{0\,RI}(\hat{t})\simeq\frac{\epsilon_{RI}}{\sqrt{\pi}\,\Gamma(3/4)}\int_0^{\hat{t}}\left[\int_0^{\hat{t}'}(\hat{t}'-\hat{t}'')^{-1/4}\,\hat{t}''^{-1/2}\,d\hat{t}''\right]d\hat{t}'.\tag{188}$$

But,

$$\int_0^{\hat{t}'}(\hat{t}'-\hat{t}'')^{-1/4}\,\hat{t}''^{-1/2}\,d\hat{t}''=\hat{t}'^{\,1/4}\int_0^1(1-x)^{-1/4}\,x^{-1/2}\,dx,\tag{189}$$

and (AS: 6.2.1, 6.2.2)

$$\int_0^1(1-x)^{-1/4}\,x^{-1/2}\,dx=\frac{\Gamma(1/2)\,\Gamma(3/4)}{\Gamma(5/4)},\tag{190}$$

so

$$\Psi_{0\,RI}(\hat{t})\simeq\frac{\epsilon_{RI}\,\hat{t}^{\,5/4}}{\Gamma(9/4)}.\tag{191}$$

At large $\hat{t}$,

$$\hat{\Psi}_{0\,RI}(\hat{t})\simeq\frac{\epsilon_{RI}\,Q_i^{1/4}}{\sqrt{2\pi}}\int_0^{\infty}{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\left[\int_0^{\hat{t}'}(\hat{t}'-\hat{t}'')^{-1/4}\,\hat{t}''^{-1/4}{\rm e}^{\,{\rm i}\,(Q_e-Q_i/2)\,\hat{t}''}\,J_{-1/4}\left(\frac{Q_i}{2}\,\hat{t}''\right)d\hat{t}''\right]d\hat{t}',\tag{192}$$

so

$$\hat{\Psi}_{0\,RI}(\hat{t})\simeq\frac{\epsilon_{RI}\,Q_i^{1/4}}{\sqrt{2\pi}}\int_0^{\infty}\left[\int_{\hat{t}''}^{\infty}(\hat{t}'-\hat{t}'')^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}'\right]\hat{t}''^{-1/4}\,{\rm e}^{\,{\rm i}\,(Q_e-Q_i/2)\,\hat{t}''}\,J_{-1/4}\left(\frac{Q_i}{2}\,\hat{t}''\right)d\hat{t}''.\tag{193}$$

But,

$$\int_{\hat{t}''}^{\infty}(\hat{t}'-\hat{t}'')^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}'={\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}''}\int_0^{\infty}x^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,x}\,dx,\tag{194}$$

and (GR: 3.381.5),

$$\int_0^{\infty}x^{\nu-1}\,{\rm e}^{-{\rm i}\,q\,x}\,dx=\frac{\Gamma(\nu)}{({\rm i}\,q)^{\nu}},\tag{195}$$

so

$$\int_{\hat{t}''}^{\infty}(\hat{t}'-\hat{t}'')^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}'={\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}''}\,\frac{\Gamma(3/4)}{[{\rm i}\,(Q_E+Q_e)]^{3/4}}.\tag{196}$$

It follows that

$$\hat{\Psi}_{0\,RI}(\hat{t})=\frac{\epsilon_{RI}\,Q_i^{1/4}}{\sqrt{2\pi}}\,\frac{\Gamma(3/4)}{[{\rm i}\,(Q_E+Q_e)]^{3/4}}\int_0^{\infty}\hat{t}''^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_i/2)\,\hat{t}''}\,J_{-1/4}\left(\frac{Q_i}{2}\,\hat{t}''\right)d\hat{t}''.\tag{197}$$

But (GR: 6.623.1),

$$\int_0^\infty \hat{t}''^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_i/2)\,\hat{t}''}\,J_{-1/4}\left(\frac{Q_i}{2}\,\hat{t}''\right)d\hat{t}'' = \frac{(Q_i)^{-1/4}\,\Gamma(1/4)}{\sqrt{\pi}\,({\rm i}\,Q_E)^{1/4}\,[{\rm i}\,(Q_E+Q_i)]^{1/4}}. \tag{198}$$

Now (AS: 6.1.17),

$$\Gamma(3/4)\,\Gamma(1/4) = \sqrt{2}\,\pi, \tag{199}$$

so

$$\hat{\Psi}_{0\,RI}(\hat{t}) \simeq \frac{\epsilon_{RI}}{[{\rm i}\,(Q_E+Q_e)]^{3/4}\,({\rm i}\,Q_E)^{1/4}\,[{\rm i}\,(Q_E+Q_i)]^{1/4}}. \tag{200}$$

6 Reconnection Scenarios

6.1 Layer Widths

Let δ be the radial width of the resonant layer, and let $\hat{\delta} = S^{1/3}\,\delta/r_s$. According to the analysis of Sects. 4.2 and 4.4, $\hat{\delta} \simeq |G|^\nu$. In estimating the layer width, we shall assume that at normalized time $\hat{t}$ the dominant layer response is associated with $g \sim 1/\hat{t}$.

In the inertial response regime, we find that

$$\hat{\delta}_I(\hat{t}) = \left(\hat{t}^{-2} + Q_E^2\right)^{1/4}\left[\hat{t}^{-2} + (Q_E+Q_i)^2\right]^{1/4}. \tag{201}$$

Thus, at small times, the layer width decreases in time as $\hat{t}^{-1}$. However, this decrease eventually stalls, and the layer width attains a constant value.

In the viscous-inertial response regime, we find that

$$\hat{\delta}_{VI}(\hat{t}) = P_\varphi^{1/4}\left[\hat{t}^{-2} + (Q_E+Q_i)^2\right]^{1/8} \tag{202}$$

Thus, at small times, the layer width decreases in time as $\hat{t}^{-1/4}$. However, this decrease eventually stalls, and the layer width attains a constant value.

In the diffusive-inertial response regime, we find that

$$\hat{\delta}_{DI}(\hat{t}) = \left(\frac{\iota_e\,P_\perp}{D^2}\right)^{1/2}\left[\hat{t}^{-2} + (Q_E+Q_i)^2\right]^{1/4}. \tag{203}$$

Thus, at small times, the layer width decreases in time as $\hat{t}^{-1/2}$. However, this decrease eventually stalls, and the layer width attains a constant value.

In the semi-collisional response regime, we find that

$$\hat{\delta}_{SC}(\hat{t}) = D^{-1}\left(\hat{t}^{-2} + Q_E^2\right)^{1/2}. \tag{204}$$

Thus, at small times, the layer width decreases in time as $\hat{t}^{-1/2}$. However, this decrease eventually stalls, and the layer width attains a constant value.

In the resistive-inertial response regime, we find that

$$\hat{\delta}_{RI}(\hat{t}) = \frac{\left(\hat{t}^{-2} + Q_E^2\right)^{1/8}\left[\hat{t}^{-2} + (Q_E + Q_i)^2\right]^{1/8}}{\left[\hat{t}^{-2} + (Q_E + Q_e)^2\right]^{1/8}}. \tag{205}$$

Thus, at small times, the layer width decreases in time as $\hat{t}^{-1/4}$. However, this decrease eventually stalls, and the layer width attains a constant value.

In the viscous-resistive response regime, we find that

$$\hat{\delta}_{VR}(\hat{t}) = P_\varphi^{1/6}. \tag{206}$$

Thus, the layer width is independent of time.

Finally, in the diffusive-resistive response regime, we find that

$$\hat{\delta}_{DR}(\hat{t}) = \left(\frac{\iota_e\,P_\perp}{D^2}\right)^{1/4}. \tag{207}$$

Thus, the layer width is independent of time.

6.2 Reconnection Scenarios 1–7

The layer widths calculated in Sect. 6.1 allow us to determine how the plasma response in the resonant layer navigates through the various linear response regimes. This navigation is illustrated in Figs. 3 and 4. Here, $\hat{t} = t/(S^{1/3}\,\tau_H)$.

Consider reconnection scenario 1, which corresponds to the horizontal lines labelled 1 in Figs. 3 and 4. This scenario holds when $D < 1$ and $P > 1$, or when $D > 1$ and $P > D^6$. The plasma in the resonant layer starts off in the inertial regime, passes into the visco-inertial regime if and when $\hat{\delta}_I(\hat{t}) = \hat{\delta}_{VI}(\hat{t})$, and passes into the viscous-resistive regime if and when $\hat{\delta}_{VI}(\hat{t}) = \hat{\delta}_{VR}(\hat{t})$. Thus, the normalized reconnected magnetic flux is specified by $\hat{\Psi}_{0\,I}(\hat{t})$ in the first time interval, by $\hat{\Psi}_{0\,VI}(\hat{t})$ in the second interval, and by $\hat{\Psi}_{0\,VR}(\hat{t})$ in the third. Analogous statements can be made for the other reconnection scenarios.

Consider reconnection scenario 2, which corresponds to the horizontal line labelled 2 in Fig. 3. This scenario holds when $D < 1$ and $D^6 < P < 1$. The plasma in the resonant layer starts off in the inertial regime, passes into the resistive-inertial regime if and when $\hat{\delta}_I(\hat{t}) = \hat{\delta}_{RI}(\hat{t})$, and passes into the viscous-resistive regime if and when $\hat{\delta}_{RI}(\hat{t}) = \hat{\delta}_{VR}(\hat{t})$.

Consider reconnection scenario 3, which corresponds to the horizontal line labelled 3 in Fig. 3. This scenario holds when $D < 1$ and $P < D^6$. The plasma in the resonant layer starts off in the inertial regime, passes into the resistive-inertial regime if and when $\hat{\delta}_I(\hat{t}) = \hat{\delta}_{RI}(\hat{t})$, passes into the semi-collisional regime if and when $\hat{\delta}_{RI}(\hat{t}) = \hat{\delta}_{SC}(\hat{t})$, and passes into the viscous-resistive regime if and when $\hat{\delta}_{SC}(\hat{t}) = \hat{\delta}_{RI}(\hat{t})$.

Consider reconnection scenario 4, which corresponds to the horizontal line labelled 4 in Fig. 4. This scenario holds when $D > 1$ and $D^3 < P < D^6$. The plasma in the resonant layer starts off in the inertial regime, passes into the viscous-inertial regime if and when

$\hat{\delta}_I(\hat{t}) = \hat{\delta}_{VI}(\hat{t})$, passes into the diffusive-inertial regime if and when $\hat{\delta}_{VI}(\hat{t}) = \hat{\delta}_{DR}(\hat{t})$, and passes into the diffusive-resistive regime if and when $\hat{\delta}_{DI}(\hat{t}) = \hat{\delta}_{DR}(\hat{t})$.

Consider reconnection scenario 5, which corresponds to the horizontal line labelled 5 in Fig. 4. This scenario holds when $D > 1$ and $D^2 < P < D^5$. The plasma in the resonant layer starts off in the inertial regime, passes into the diffusive-inertial regime if and when $\hat{\delta}_I(\hat{t}) = \hat{\delta}_{DI}(\hat{t})$, and passes into the diffusive-resistive regime if and when $\hat{\delta}_{DI}(\hat{t}) = \hat{\delta}_{DR}(\hat{t})$.

Consider reconnection scenario 6, which corresponds to the horizontal line labelled 6 in Fig. 4. This scenario holds when $D > 1$ and $D^{-6} < P < D^2$. The plasma in the resonant layer starts off in the inertial regime, and passes into the diffusive-resistive regime if and when $\hat{\delta}_I(\hat{t}) = \hat{\delta}_{DR}(\hat{t})$.

Finally, consider reconnection scenario 7, which corresponds to the horizontal line labelled 7 in Fig. 4. This scenario holds when $D > 1$ and $P < D^{-6}$. The plasma in the resonant layer starts off in the inertial regime, passes into the semi-collisional regime if and when $\hat{\delta}_I(\hat{t}) = \hat{\delta}_{SC}(\hat{t})$, and passes into the diffusive-resistive regime if and when $\hat{\delta}_{SC}(\hat{t}) = \hat{\delta}_{DR}(\hat{t})$.

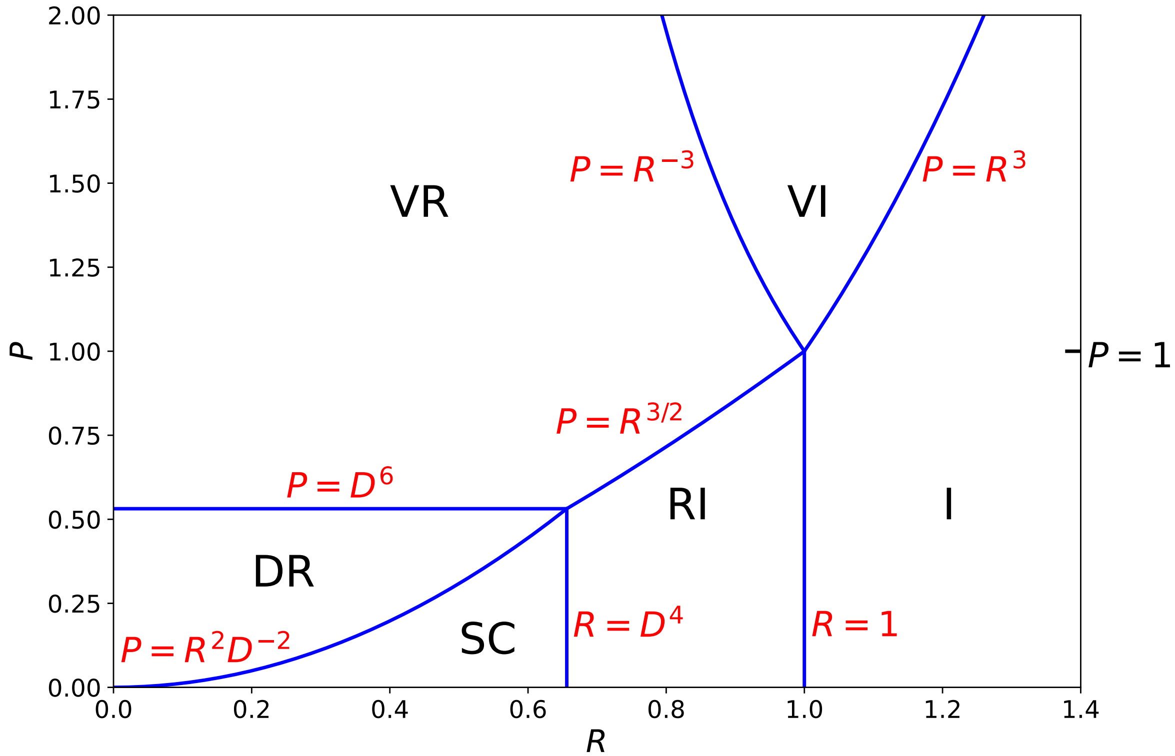

Figure 1: Linear resonant plasma response regimes in R-P space for the case $D = 0.9$. The various regimes are the diffusive-resisitive (DR), the semi-collisional (SC), the resistive-inertial (RI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).

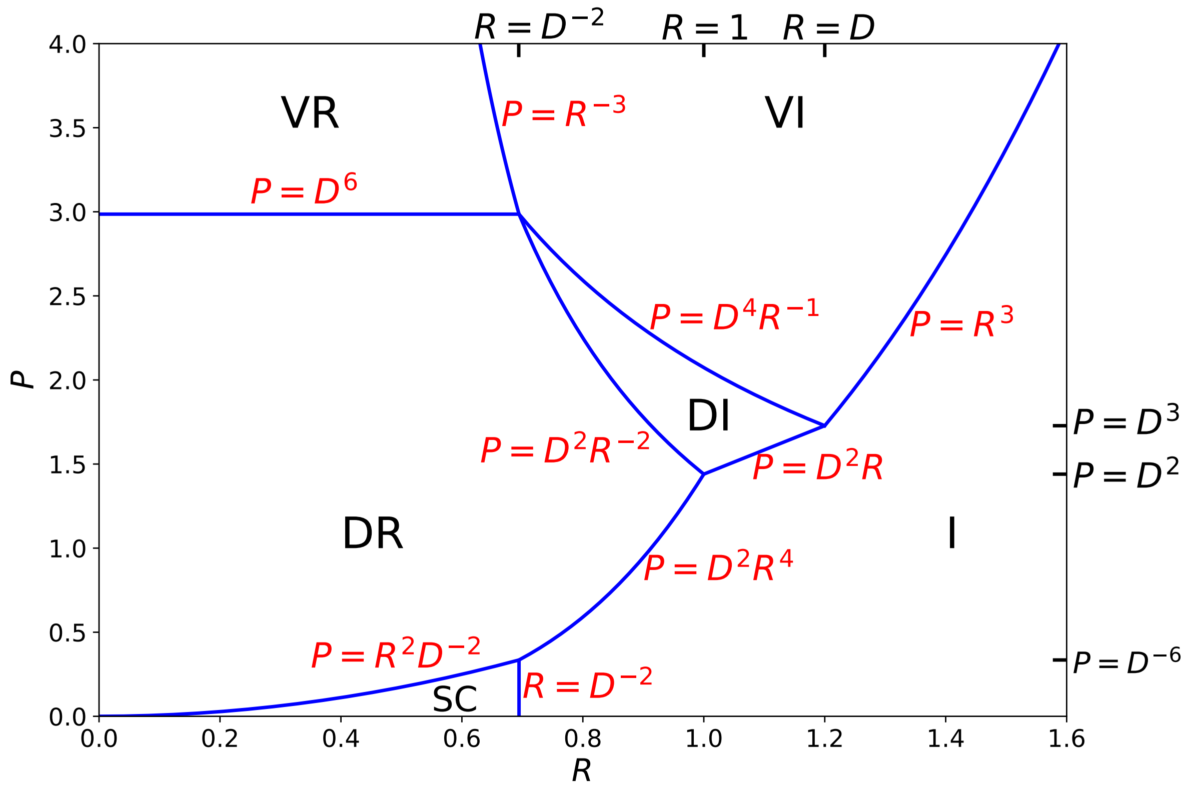

Figure 2: Linear resonant plasma response regimes in R-P space for the case $D = 1.2$. The various regimes are the diffusive-resisitive (DR), the semi-collisional (SC), the diffusive-inertial (DI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).

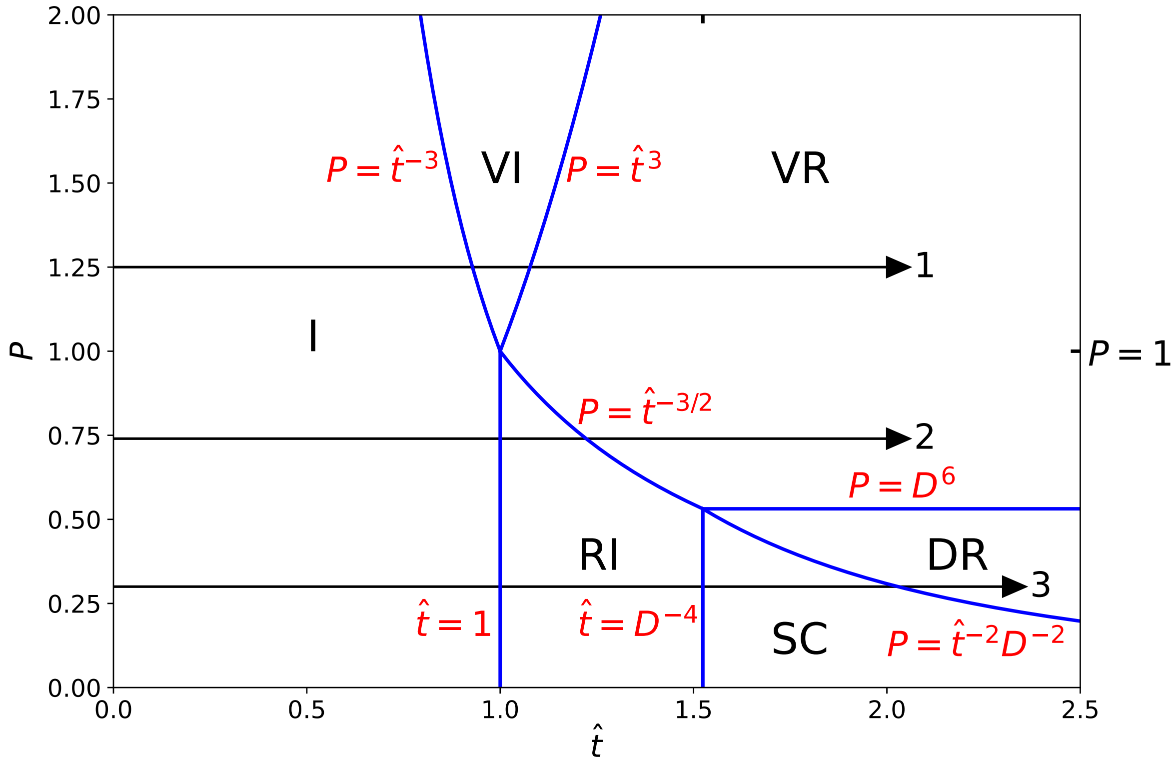

Figure 3: Trajectory of the Fourier/Laplace transformed response of a resonant layer to a resonant magnetic perturbation in $\hat{t}$-P space for the case $D = 0.9$. The various regimes are the diffusive-resistive (DR), the semi-collisional (SC), the resistive-inertial (RI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).

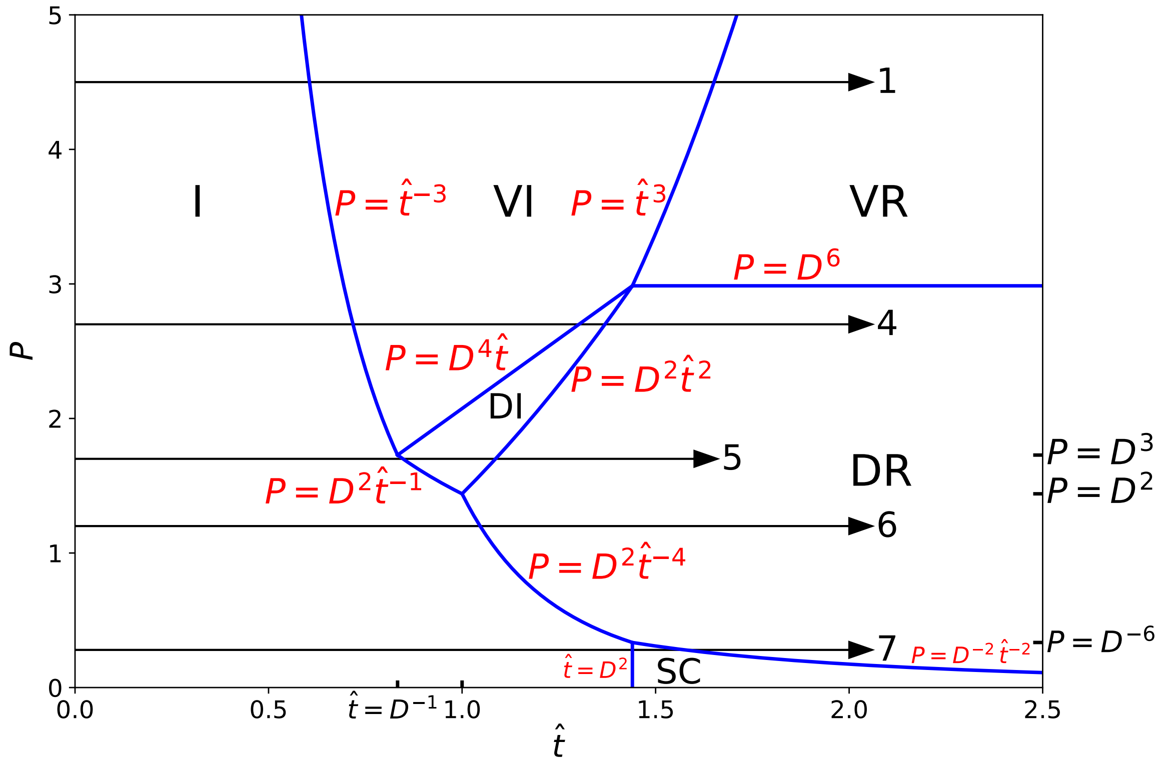

Figure 4: Trajectory of the Fourier/Laplace transformed response of a resonant layer to a resonant magnetic perturbation in $\hat{t}$-P space for the case $D = 1.2$. The various regimes are the diffusive-resistive (DR), the semi-collisional (SC), the diffusive-inertial (DI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).